

DSpi3 Build guide

By William Oprey :)

Please note:

This is an intermediate to difficult level project. To build this, you must know how to solder at minimum, and there is some fiddly work that may be difficult for some people. You also need either an FDM or Resin printer to produce parts, and having an FDM printer is useful for the screen removal as well.

I haven't included any instructions for this, but I highly recommend some post processing + painting of the main parts before assembly, as it looks quite nice.

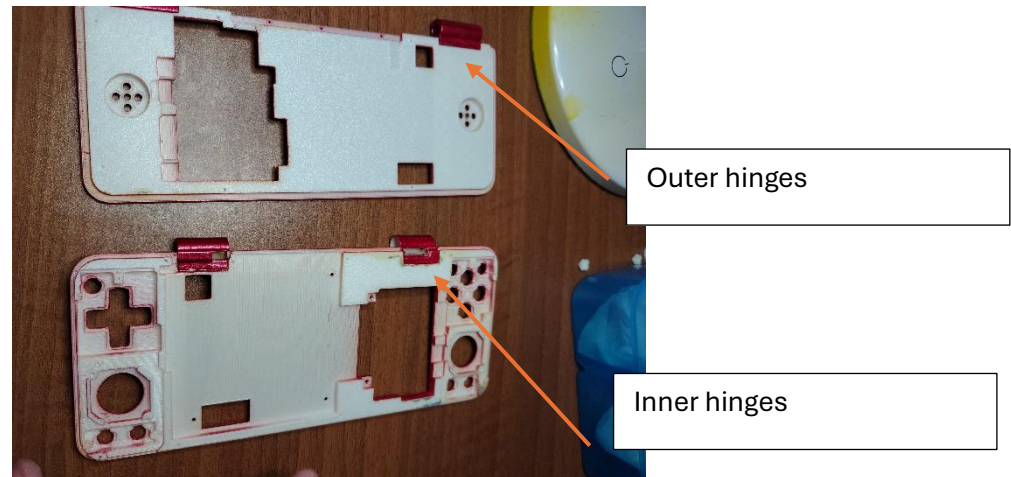
Parts list:

Part	Quantity
RasPI 5 Compute module lite	1
DSPI Motherboard	1
DSPI Left controller board	1
DSPI Right controller board	1
Bottom Plate Print	1
Bottom Cover Print	1
Lower Hinge Right Print	1
Lower Hinge Left Print	1
Lower Hinge Middle Print	1
Fan Guide Print	1
Battery Holder Print	1
Top Plate Print	1
Top Cover Print	1
Top Hinge Left Print	1
Top Hinge Right Print	1
PCB Inserts Print	1
Buttons Prints	1
Waveshare 5-DSI-TOUCH-A screen	2
GBA SP Hinge	2
Soberton SP-1605 Speakers	2
3DS Thumbstick (inc cap)	2
Raspberry pi 5 fan 30x30	1
Pi CM5 low-profile heatsink	1
5000maH 955565 LiPo battery	1
15cm 22p same side ribbon cable	1
30cm 22p opposite side ribbon cable	1
10cm 22p same side ribbon cable	1
10cm 22p opposite side ribbon cable	1
JST PH 2.0 2p cable	2
JST PH 1.25 2p cable	2
JST SUR 0.8 2p cable	2
M1.7x4 machine screw	20
M1.7x8 self-taping screw	4
M1.7x3 machine screw	11
M2x5 machine screw	4
6x6x3.1 tactile pushbutton	15
JST 2.0 header right angle	2

Build steps:

Step 1:

Apply super glue to the hinge cutouts on the top and bottom plates, then attach the outer and inner hinges to their respective plates:

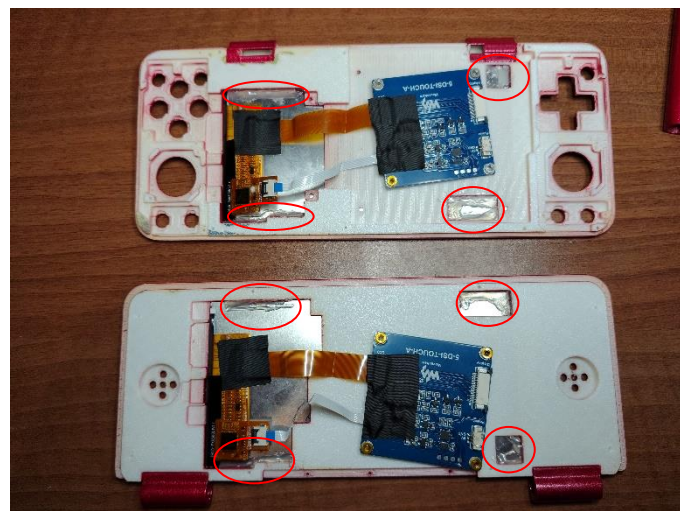


Step 2:

Remove the screens from the metal enclosures they come in, steps for this are in a separate file on the GitHub

Step 3:

Insert the screens into the top and bottom plates, then apply hot glue into the cutouts provided. Ensure the hot glue does not protrude out from the plate, so the pcbs mount correctly later.



Step 4:

Insert the GBA hinges into the outer hinge on the top plate. Ensuring the hinge is aligned to the guides in the body, such as in the image below:



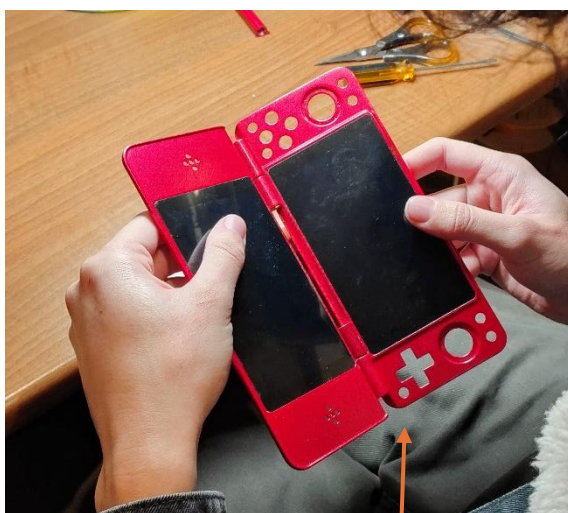
Ensure guide is aligned with slot on hinge

Push the hinge until it is in the position shown in the image below. Repeat for both sides.

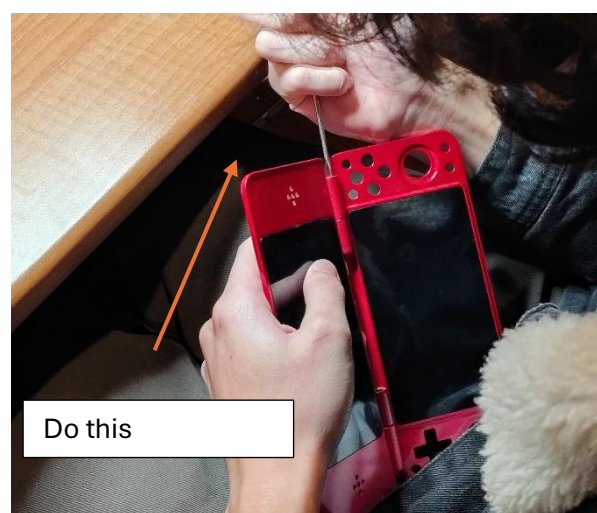


Step 5:

Push the hinges back until they are no longer protruding from the top plate, then place the top plate over the bottom plate, at an angle 30 degrees up from flat. Once in this position, use a screwdriver to push both hinges into place. To help with alignment, the correct position is where the guides on the top and bottom plate hinge are aligned.



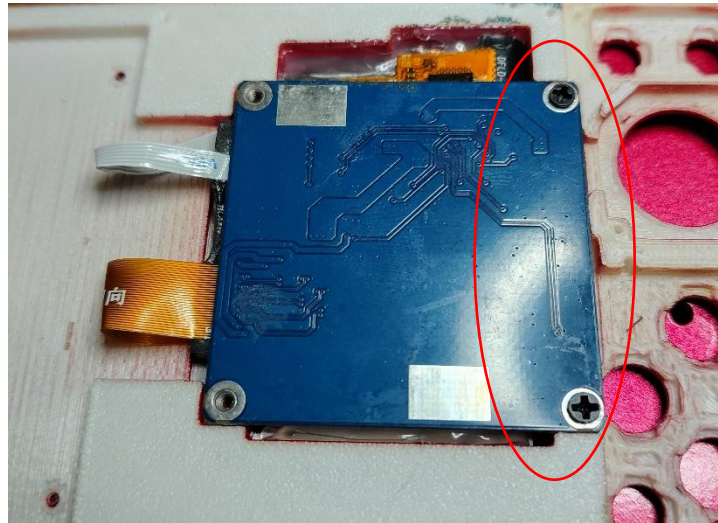
Hold at this angle



Do this

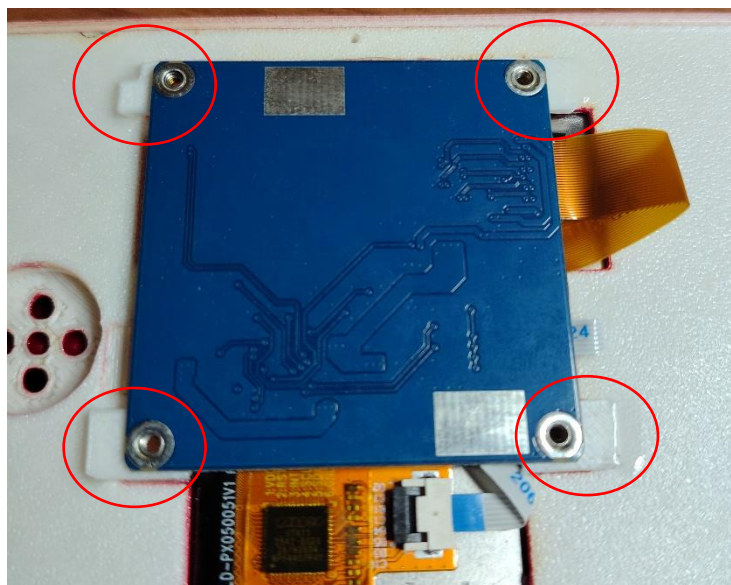
Step 5:

Secure the screen PCB to the bottom plate using the original screws from the screen case. Only screw in the screws shown in the picture below.



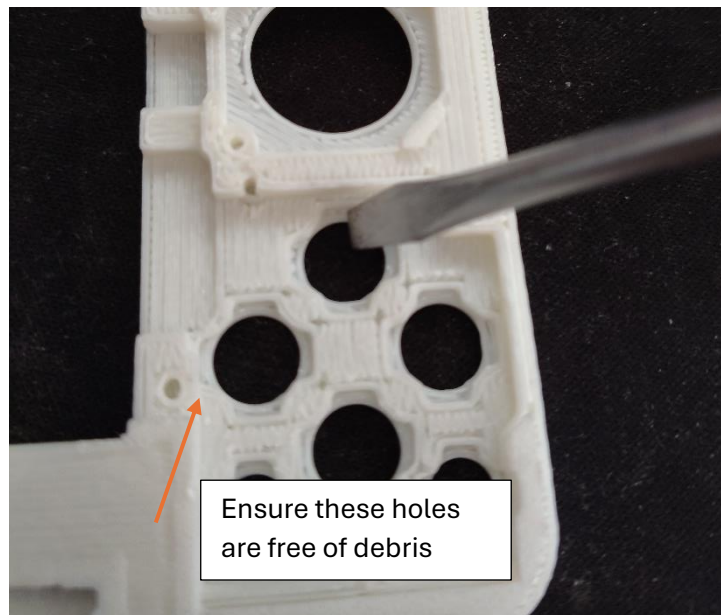
Step 6:

Get the screen PCB inserts (3D Print) and place them over the standoffs on the upper screen PCB. Place the PCB over the top plate with inserts in the provided slots, then superglue the inserts into place.



Step 7:

Place the front facing buttons into the plate. Ensure that all buttons go fully into place, if any get stuck before this you may have to use a screwdriver to clear out the button guide tabs.



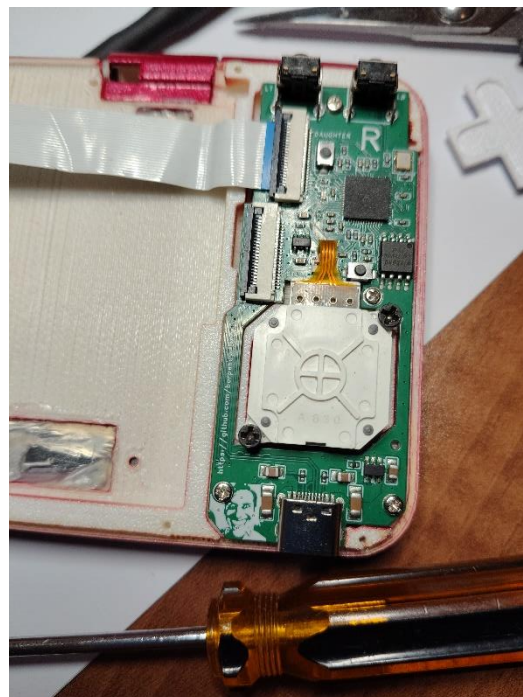
Step 8:

Solder the buttons to the controller PCBs. Ensure that you solder the bumper buttons first, then use clippers to remove any protruding pins, before soldering the face buttons.

Step 9:

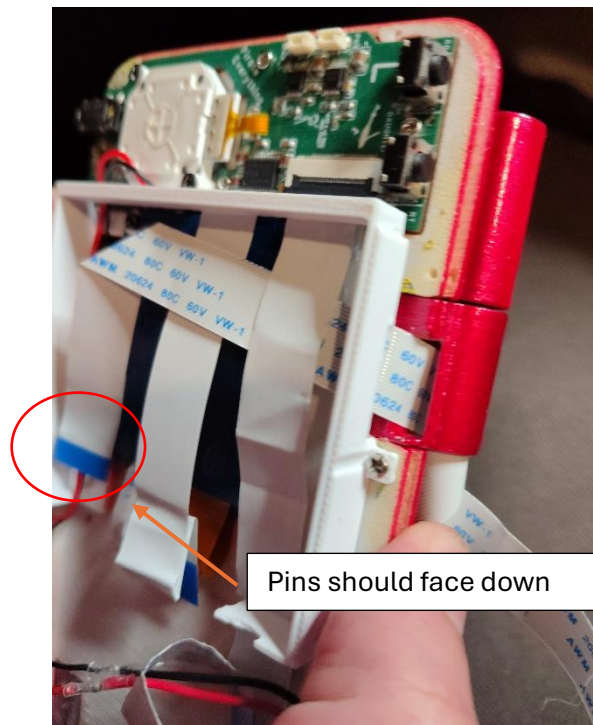
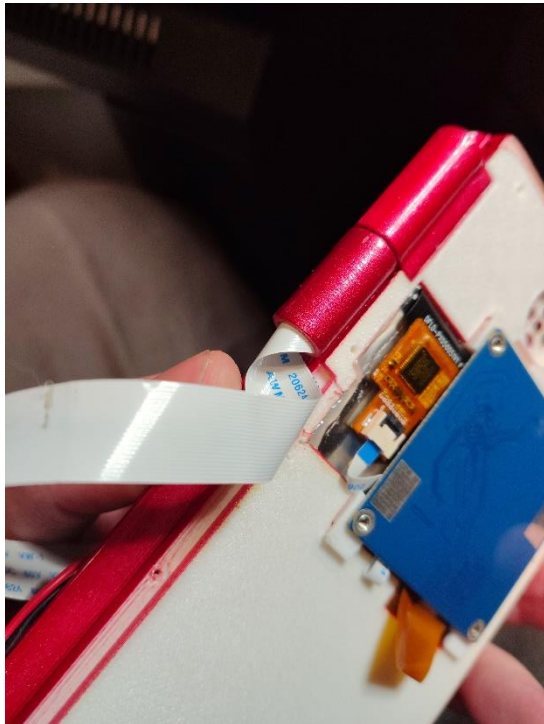
Screw in the left and right controller PCBs to the bottom plate. Test all the buttons once PCBs are inserted, there should be a click sound and feel. Insert the thumb stick FPC cables into their connectors and screw the thumb sticks to the plate. Connect the PCBs together using a 22p 15cm same side ribbon cable.

Use m1.7x4 machine screws for the PCBs and m1.7x8 self-taping screws for the sticks



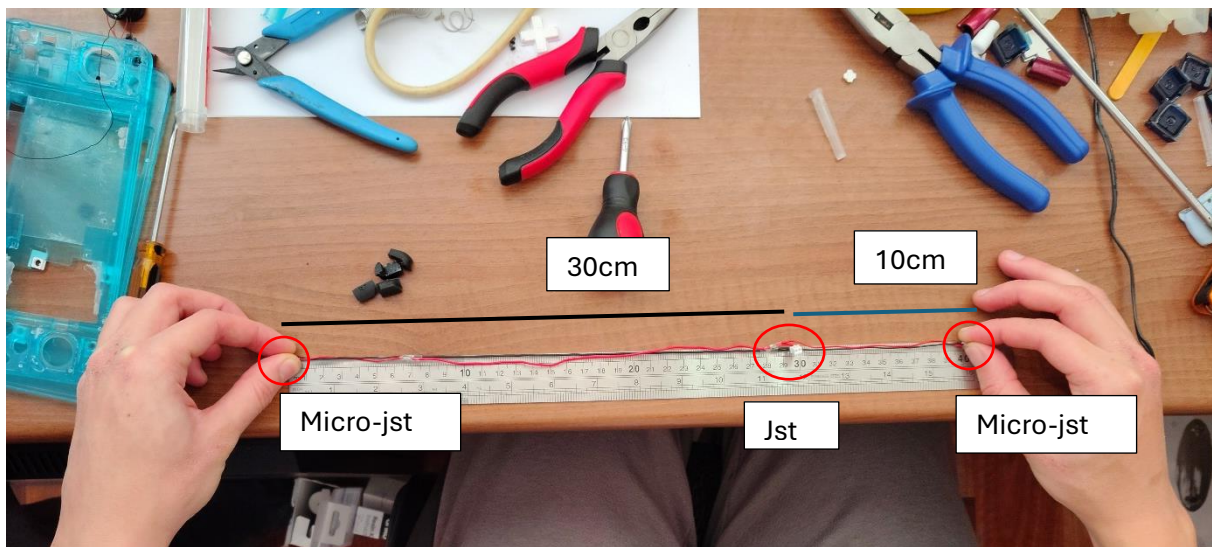
Step 10:

Roll a 30cm 22p different side ribbon cable through the left hinge, using the provided hole. Ensure the orientation of the cable matches the one pictured



Step 11:

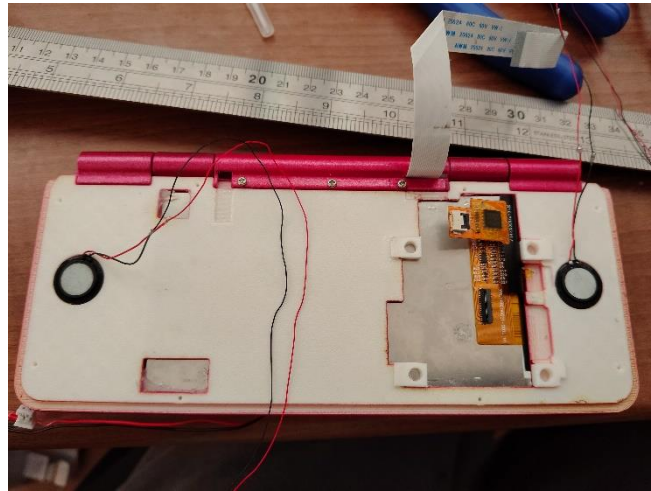
Build the screen power cable to match the one pictured below. The cable is a jst2.5 to 2xmicro-jst1.25 y adapter. One side of the cable is 30cm, and the other is 10cm long. Use the micro jst cables provided with the screens.



Step 12:

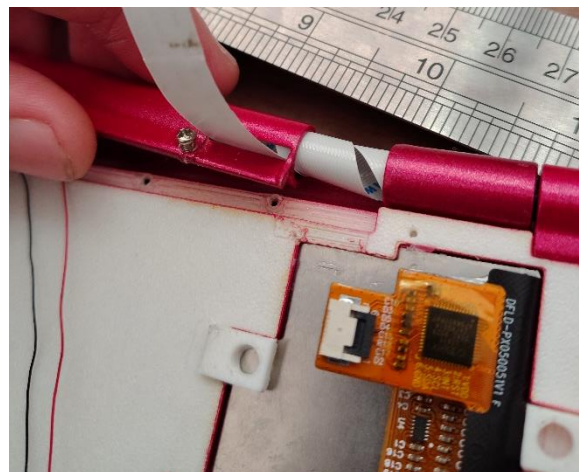
Build the speaker assembly by soldering the JST SUR 0.8 cables to the speakers. The left speaker wire length should be 24cm, and the right speaker wire length should be 30cm.

To achieve this, you may need to shorten the left wire and lengthen the right wire. Then, place the speakers into the top plate.



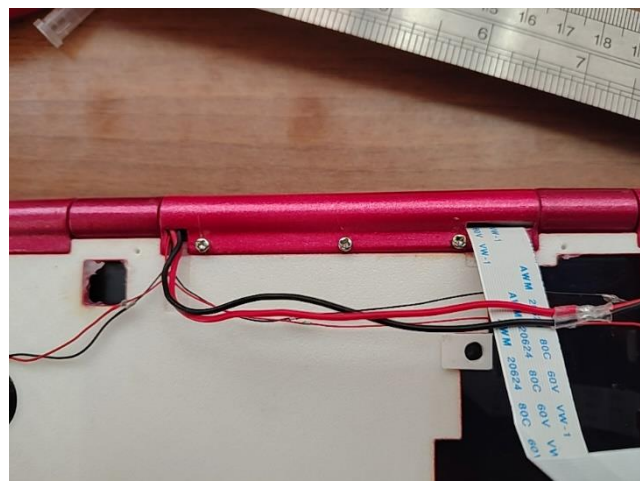
Step 13:

Roll the ribbon cable through the middle hinge segment, then screw the middle hinge into the top plate. Use m1.7x3 screws.



Step 14:

Run the longer end of the power cable through the provided hole on the left of the hinge, then run the speaker cables through the same hole.



Step 15:

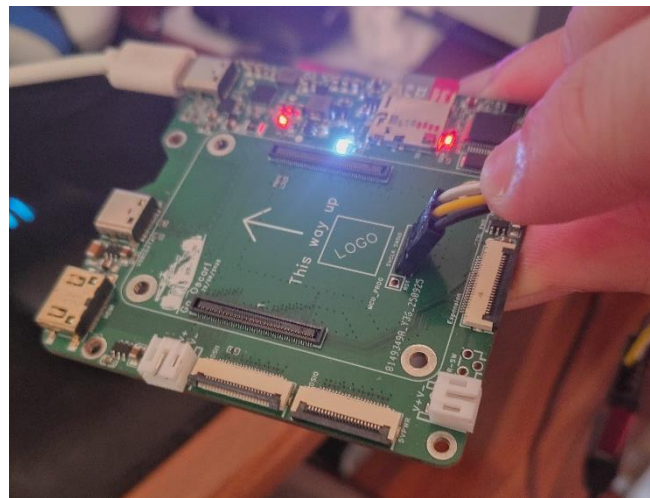
Connect the power and DSI ribbon cable to the upper screen PCB.

Step 16:

Solder the JST connectors for the battery and screens onto the DSpi motherboard. After soldering, trim off the pins on the bottom of the board.

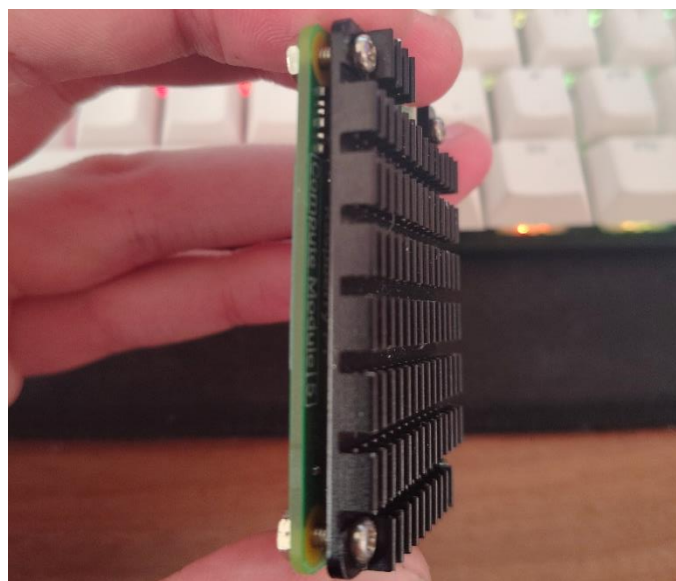
Step 17:

Upload the firmware onto the DSpi motherboard using a st-link, and the programming pinholes on the board. Make sure the board is connected to power before programming. The firmware can be found in the GitHub page, use stm32cubeide to program.



Step 18:

Apply thermal paste to the CPU on the compute module, then attach the heat sink. To reduce the height of the heatsink, mount it using the provided screws, as shown in the below image:



Step 19:

Superglue the cooling fan to the 3D printed fan guide. To ensure the fan cable can reach the motherboard, unclip the cable from the fan housing, and route it underneath the guide

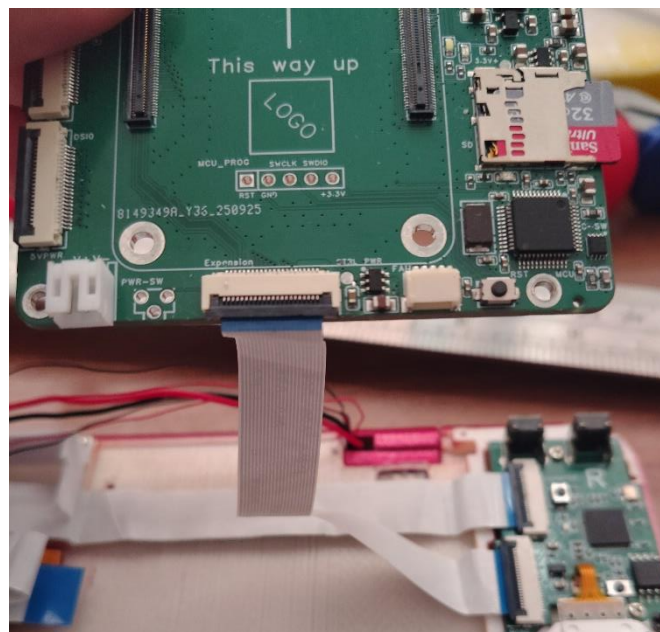


Step 20:

Program a SD card with the provided OS, or a fresh OS install, then insert the SD card into the motherboard.

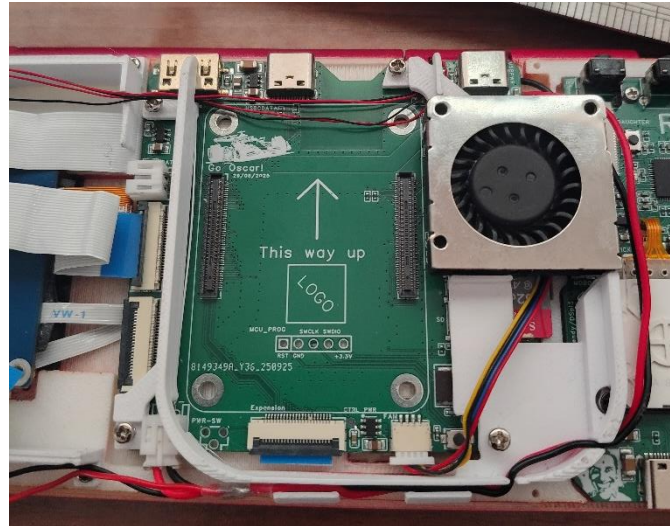
Step 21:

Connect the motherboard to the right controller using a 10cm same side 22p ribbon cable.



Step 22:

Place the motherboard on the bottom plate, making sure the power cable sits below in the provided hinge. Place the fan guide over the motherboard, then route the power cable through the clips, and plug it into the motherboard



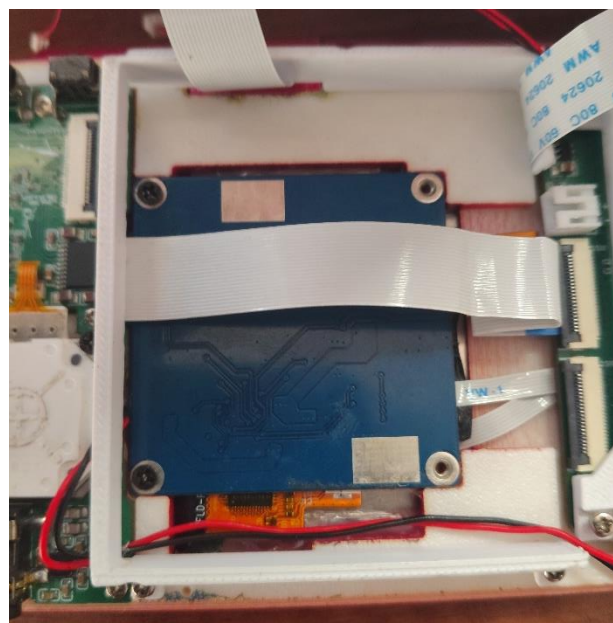
Step 23:

Screw in the motherboard to the bottom plate, making sure the screws go through the fan guide. Plug the compute module into the motherboard, making sure the antenna cutout is facing up (important).

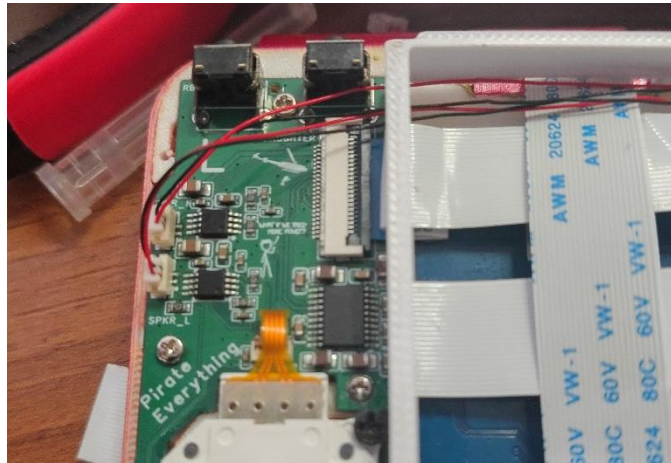
Use m2x5 machine screws, and screw until you feel some resistance then stop (Important). This is to make sure you don't screw into the screen.

Step 24:

Unscrew the bottom screen PCB, then plug in the power cable, and connect the display to the motherboard using a 10cm 22pin different side ribbon cable. Then re-attach the PCB.



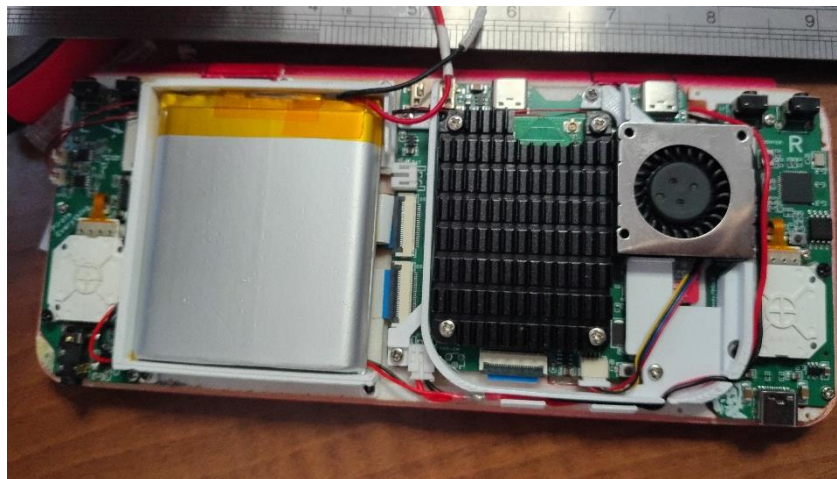
Step 25: Plug the speaker cables into the connectors on the left controller PCB.



Step 25:

Screw the battery holder into the bottom plate, then place the battery into the holder and connect it to the motherboard. Check that the system runs (fan should start up), and that both screens display before continuing.

Use m1.7x4 machine screws.



Fully assembled bottom end should look like this

Step 26:

Place the rear bumpers into the bottom cover, then attach the bottom shell and screw it in. Once finished, check that all the rear bumper buttons work (you should feel a click).

Use m1.7x4 machine screws.



Step 27:

Screw in the top cover, making sure the top screen pcb is fully down

Use m1.7x3 machine screws.

Step 28: Follow the instructions on the GP2040-CE website to program the controller. Test that the controller works, and test that the speakers work.